QAOA for the MaxCut Problem on Weighted Random Graphs

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Review of Variational Quantum Algorithms (VQAs)

- Goal: Find the quantum state with lowest (or highest) eigenvalue of a certain observable.
- Consider a Hermitian operator H, ground state $|\psi_0\rangle$ with energy E_0.
- Method: Choose a parametrized trial state $|\psi(\vec{\theta})\rangle = U(\vec{\theta})\,|\psi_{\text{initial}}\rangle$ for $U(\vec{\theta}) = U_N(\theta_N)\ldots U_1(\theta_1)$, with $\vec{\theta} = (\theta_1, \ldots, \theta_N)^T$ and initial state $|\psi_{\text{initial}}\rangle$ where in general

$$E_0 = \langle\psi_0|\, H\, |\psi_0\rangle \leq \langle\psi(\vec{\theta})|\, H\, |\psi(\vec{\theta})\rangle \tag{1}$$

for which we hope that

$$\min_{\vec{\theta}} \langle\psi(\vec{\theta})|\, H\, |\psi(\vec{\theta})\rangle \gtrsim E_0. \tag{2}$$

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QAOA - Construction of the Circuit

- Main ingredient of the circuit: Suzuki-Trotter decomposition of the time evolution operator.
- In general, several optimization problems can be mapped on an Ising Hamiltonian

$$H(s_1, \ldots, s_N) = -\sum_{i<j} J_{ij} s_i s_j - \sum_{i=1}^{N} h_i s_i, \qquad s_i \in \{\pm 1\}, \tag{3}$$

  seeking to solve $\min_{\mathbf{s}} \langle H(\mathbf{s}) \rangle$ to find the ground state (energy).
- For the implementation on a quantum circuit, consider a quantum Ising Hamiltonian

$$H_C = -\sum_{i<j} J_{ij} Z_i Z_j - \sum_{i=1}^{N} h_i Z_i, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{4}$$

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- Also consider the **Mixer Hamiltonian**

$$H_M = -\sum_{i=1}^{N} X_i, \qquad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{5}$$

- Then set the time-dependent Hamiltonian

$$H(t) = (1 - t/T)H_M + (t/T)H_C, \qquad H(0) = H_M \text{ and } H(T) = H_C$$

for which the Trotterization of its time-evolution operator is (roughly)

$$e^{-iH\Delta t} \approx e^{-i\gamma H_C} e^{-i\beta H_M}, \qquad \beta = (1 - t/T)\Delta t \text{ and } \gamma = (t/T)\Delta t.$$

- Generalize with a non-uniform time function $g(t)$, at a depth p, for some angles γ_i and β_i, $i = \{1, \ldots, p\}$.

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QAOA for MaxCut Problem on Weighted Random Graphs

- Problem Statement: Divide the nodes of an undirected graph into two sets such that the sum of the weights is maximized.
- Mathematically: Consider an undirected graph $G = (V, E)$ with edge weights $\{w_{uv}\}_{u,v}$ associated with each edge $(u, v) \in E$. Find $\mathbf{s} \in \{\pm 1\}^{|V|}$ giving us

$$\max_{\mathbf{s}} C(\mathbf{s}) := \max_{\mathbf{s}} \sum_{(u,v)\in E} \frac{w_{uv}}{2}(1 - s_u s_v) \equiv \max_{\mathbf{s}} \sum_{(u,v)\in E} (-s_u s_v w_{uv})$$
$$\equiv -\min_{\mathbf{s}} \sum_{(u,v)\in E} s_u s_v w_{uv}.$$

- Encode MaxCut on qubits via the cost and mixer hamiltonians

$$H_C = -\sum_{(u,v)\in E} w_{uv} Z_u Z_v \quad \text{and} \quad H_M = \sum_{u\in V} X_u. \tag{6}$$

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Qiskit Implementation - Setup

- Initial state $|\psi_{\text{initial}}\rangle = (H|0\rangle)^{\otimes N} = |+\rangle^{\otimes N}$ (ground state of H_M).
- For a set of parameters $\vec{\theta} = (\vec{\gamma}, \vec{\beta}) = (\gamma_1, \ldots, \gamma_p, \beta_1, \ldots, \beta_p)$, the p-layers QAOA parametrized quantum circuit is

$$U(\vec{\gamma}, \vec{\beta}) = \prod_{j=1}^{p} e^{-i\gamma_j H_C} e^{-i\beta_j H_M}. \tag{7}$$

- In this project: (i) we generate a random weighted graph, (ii) over $N = 7$ qubits (i.e. 7 nodes), and (iii) considering 2-layers quantum circuit $\Longrightarrow$ 4 parameters to estimate $(\gamma_1, \beta_1, \gamma_2, \beta_2)$.

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Qiskit Implementation - Random Weighted Graph

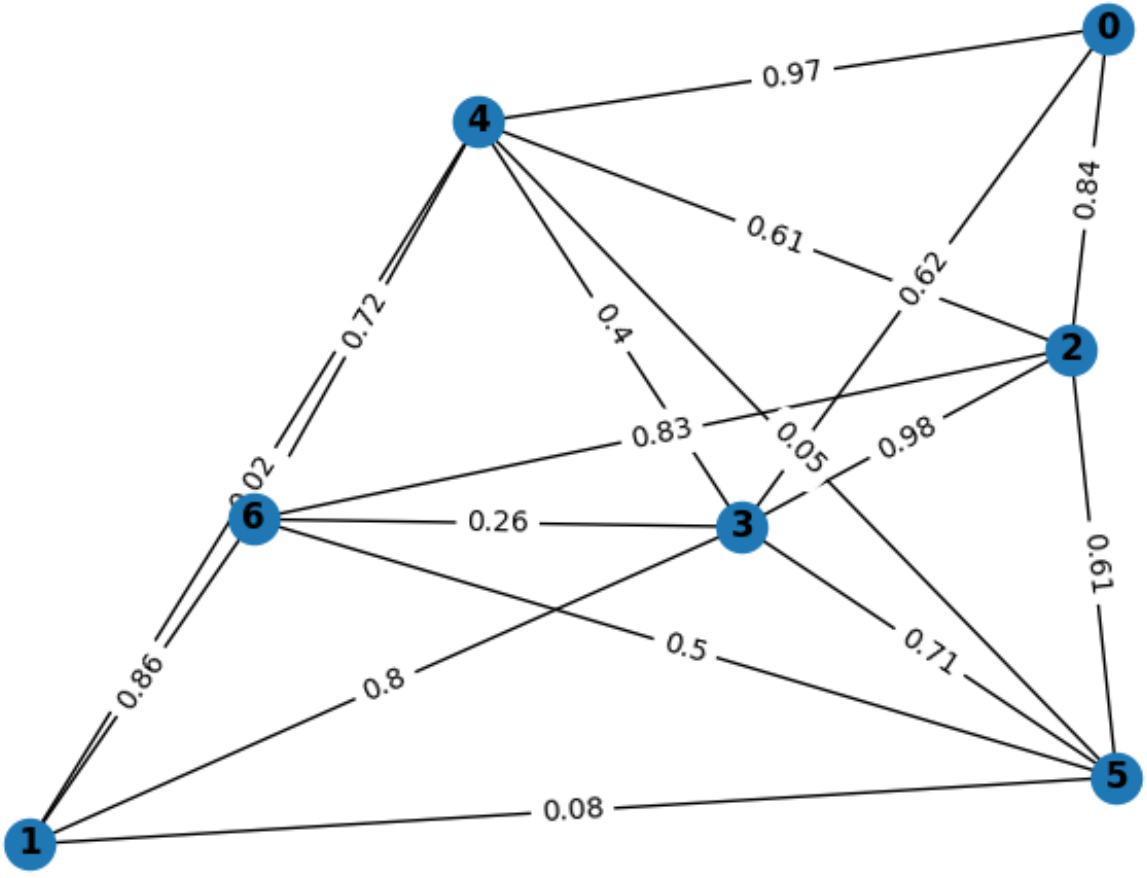


Figure: Random weighted graph with $N = 7$ nodes and 17 edges.

Qiskit Implementation - Quantum Circuit

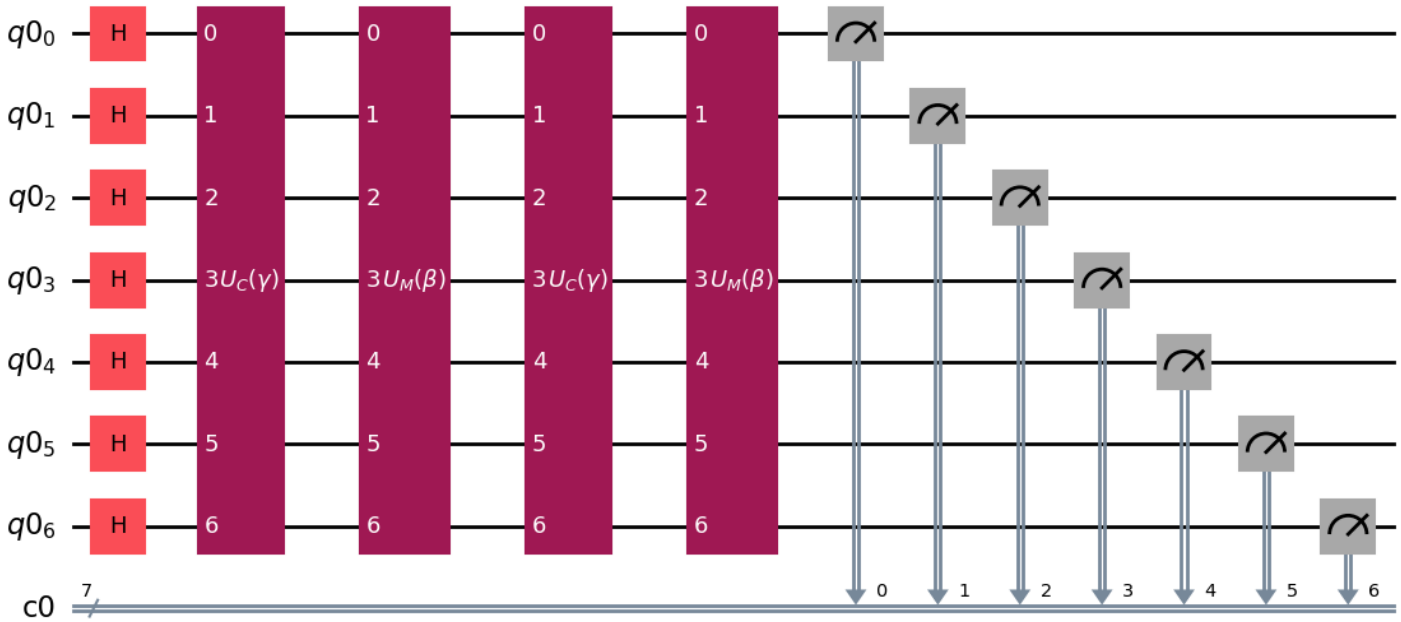


Figure: 2-layers Parametrized Quantum Circuit

Qiskit Implementation - $e^{-i\gamma H_C}$ into Gates

- For the 2-qubit unitary $e^{-iZZt} = e^{-i(Z\otimes Z)t}$ observe, for $j, k \in \{0, 1\}$,

$$\underbrace{e^{-i(Z\otimes Z)t} \left|jk\right\rangle = e^{-i(-1)^{j\oplus k}t} \left|jk\right\rangle} \tag{8}$$

Use the 1-qubit gate $R_Z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$

- Strategy: Use CNOT gate to check the parity of qubits u and v, then apply $R_Z(2\gamma w_{uv})$, and undo the effect of the first gate by another CNOT.

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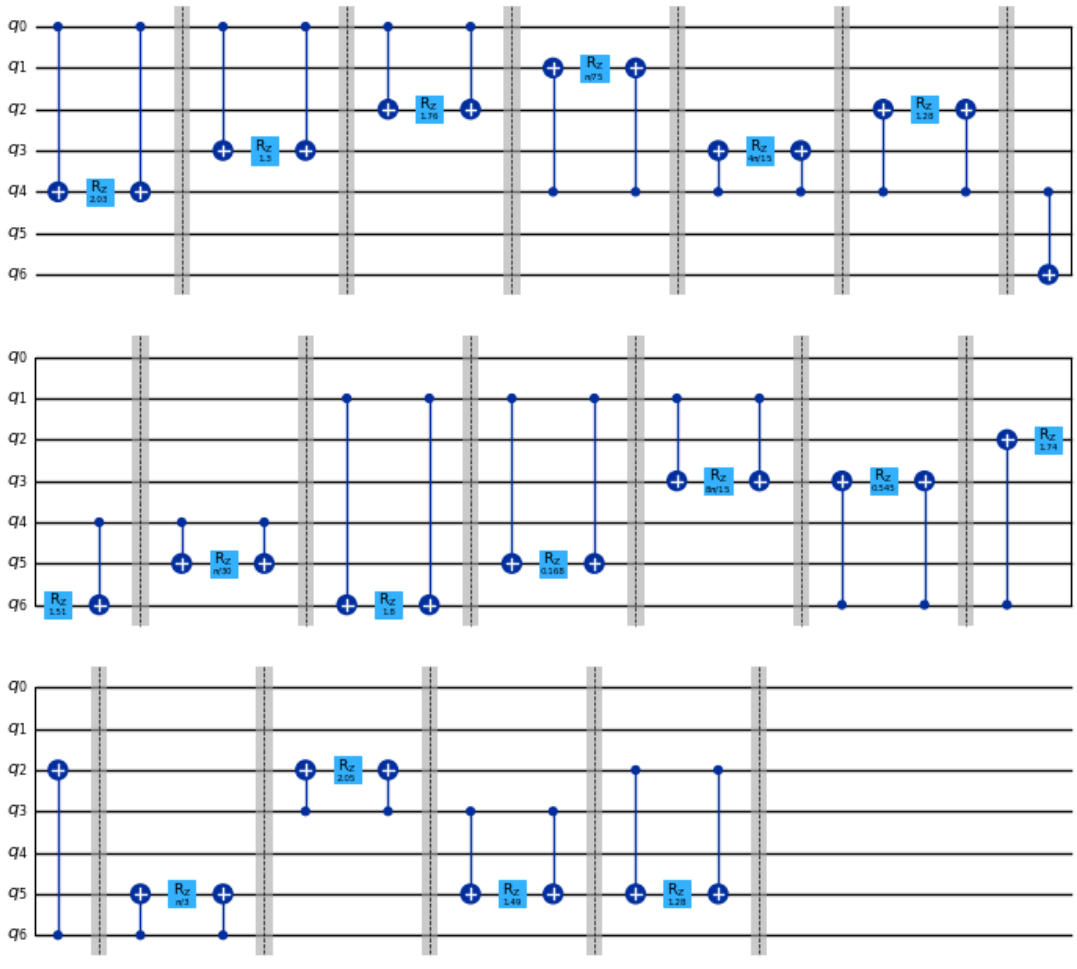
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Qiskit Implementation - $e^{-i\beta H_M}$ into Gates

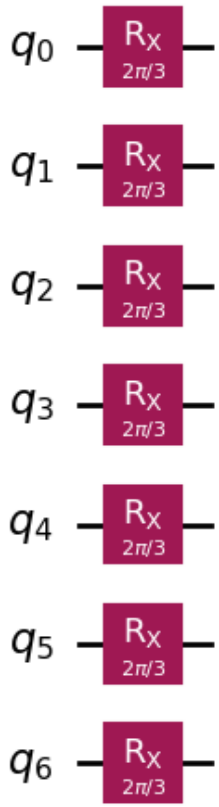


Figure: For each qubit we apply $e^{-i\beta X}$ corresponding to a $R_X(2\beta)$ gate.

Qiskit Implementation - Solving MaxCut

- Using 10 000 shots and a COBYLA method for the minimization we find that the optimal angles are

$$(\gamma_1, \beta_1, \gamma_2, \beta_2) = (2.06, 1.50, 0.60, 1.74).$$

- The corresponding value of the energy is found to be

$$\langle H_C \rangle_{\vec{\gamma},\vec{\beta}} := \langle \psi(\vec{\gamma}, \vec{\beta}) | \, H_C \, | \psi(\vec{\gamma}, \vec{\beta}) \rangle \approx 5.46$$

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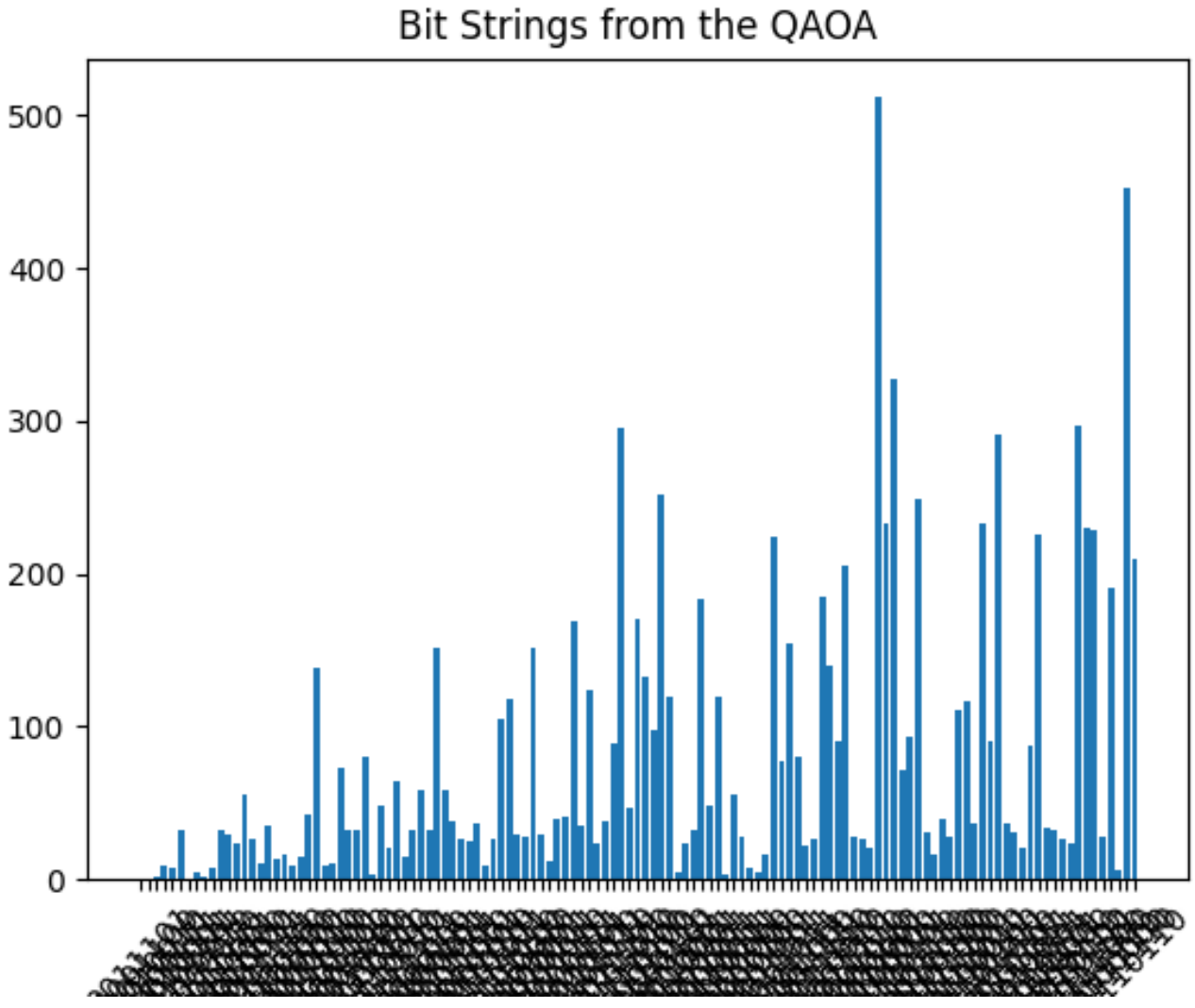
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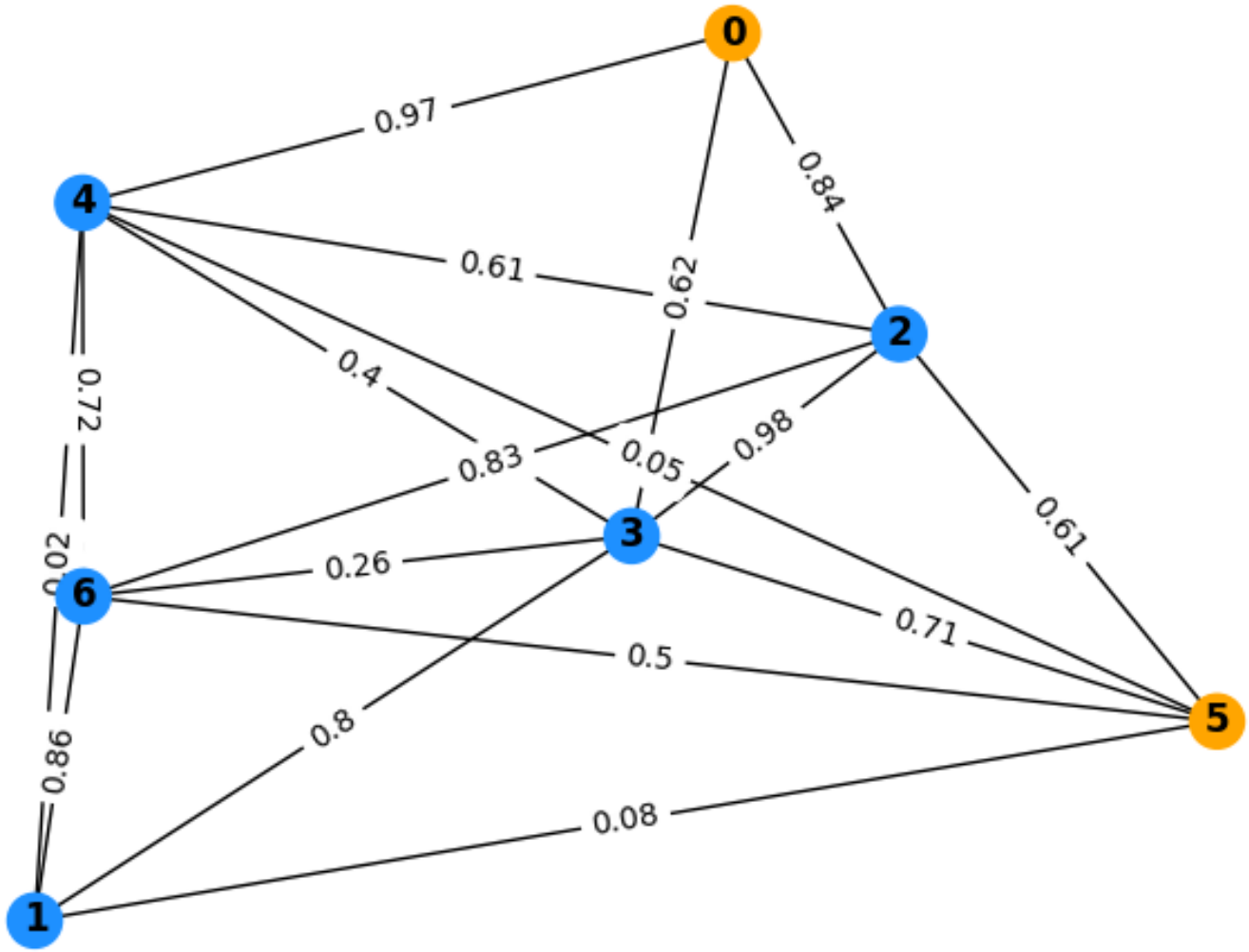
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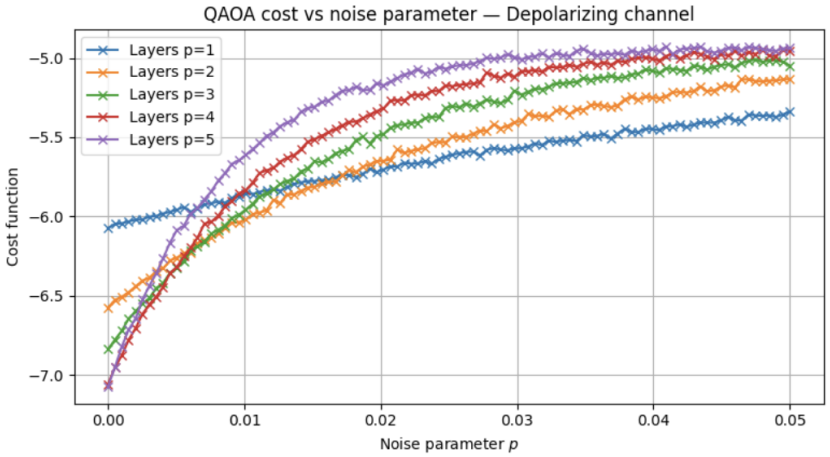
- The MaxCut bitstring is 1000001 with a cut value of 5.60.

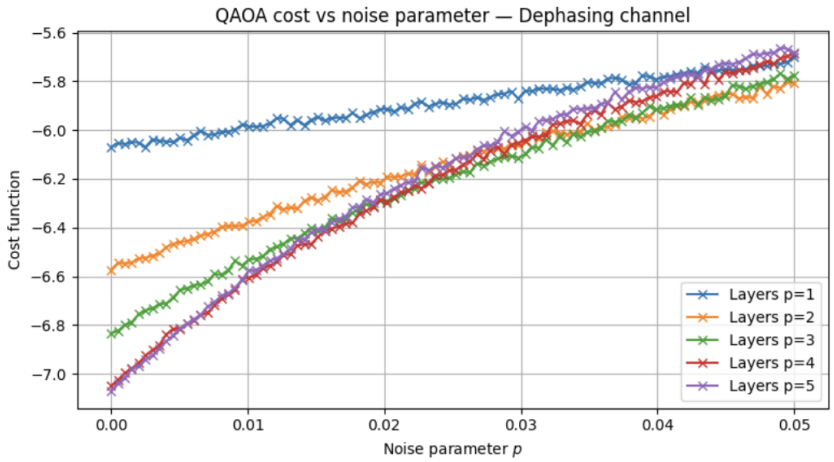


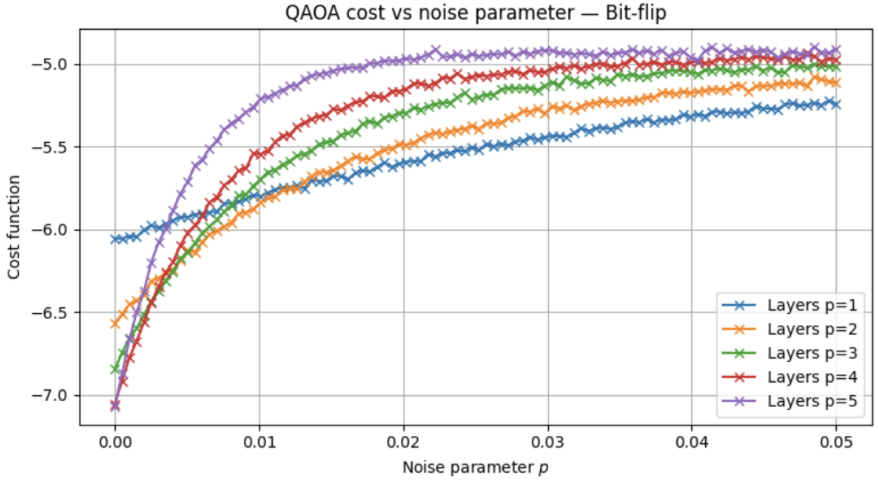
Qiskit Implementation - Solving MaxCut



Qiskit Implementation - Robustness Against Noise







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- In [5] they show that for the cost function

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  we have that

$$f(\vec{\gamma},\vec{\beta})^{\text{noisy}} \sim (1-p)^{\alpha M} f(\vec{\gamma},\vec{\beta})^{\text{ideal}} \tag{10}$$

  where $M = \#\{\text{quantum gates}\}$, $p =$ noise parameter and α a model parameter.

- Noise does not change the architecture of the parameter space but rather flattens it, with flattening factor $(1-p)^{\alpha M} \overset{p \text{ small}}{\sim} e^{-\alpha M p}$.

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[1] Suguru Endo et al. "Hybrid Quantum-Classical Algorithms and Quantum Error Mitigation". In: *Journal of the Physical Society of Japan* 90.3 (Mar. 2021), p. 032001.

[2] Tianyi Hao et al. "End-to-end protocol for high-quality quantum approximate optimization algorithm parameters with few shots". In: *Physical Review Research* 7.3 (Aug. 2025).

[3] Stefan H. Sack and Maksym Serbyn. "Quantum annealing initialization of the quantum approximate optimization algorithm". In: *Quantum* 5 (July 2021), p. 491.

[4] V.Savona. *Lecture notes on Quantum Computing*. 2025.

[5] Cheng Xue et al. *Effects of Quantum Noise on Quantum Approximate Optimization Algorithm*. 2019. arXiv: 1909.02196 [quant-ph].